

Lecture 11: Itô formula

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1 Formula for Brownian motion

We use the limit of integrals of simple integrands to obtain

$$\frac{1}{2}W^2(T) = \int_0^T W(u)dW(u) + \frac{1}{2}T. \quad (1)$$

Notice that if we let $f(x) = \frac{1}{2}x^2$, then above equality can be expressed as

$$f(W(T)) = \int_0^T f'(W(u))dW(u) + \int_0^T f''(W(u))du. \quad (2)$$

This is true in general. In particular, consider a function $f(t, x)$ which has continuous partial derivatives $f_t := \frac{\partial f}{\partial t}$, $f_x := \frac{\partial f}{\partial x}$, $f_{xx} := \frac{\partial^2 f}{\partial x^2}$, then $f(T, W(T))$ can be expressed as a stochastic integral.

To do so, consider a partition $\Pi = \{t_0, t_1, \dots, t_N\}$:

$$0 = t_0 < t_1 < \dots < t_N = T.$$

Then we trivially have

$$f(T, W(T)) - f(0, W(0)) = \sum_{i=1}^N [f(t_i, W(t_i)) - f(t_{i-1}, W(t_{i-1}))]. \quad (3)$$

Assuming the partition is fine enough and all the increments $t_i - t_{i-1}$, $W(t_i) - W(t_{i-1})$, $1 \leq i \leq N$ are sufficiently small¹ so that we can apply Taylor expansion for individual term in (3). We then have

$$\begin{aligned} & \sum_{i=1}^N [f(t_i, W(t_i)) - f(t_{i-1}, W(t_{i-1}))] \\ = & \sum_{i=1}^N \left[f_t(t_{i-1}, W(t_{i-1}))(t_i - t_{i-1}) + f_x(t_{i-1}, W(t_{i-1}))(W(t_i) - W(t_{i-1})) \right. \\ & \left. + \frac{1}{2}f_{xx}(t_{i-1}, W(t_{i-1}))(W(t_i) - W(t_{i-1}))^2 + \text{terms proportional to } (t_i - t_{i-1})^p(W(t_i) - W(t_{i-1}))^q \right], \end{aligned}$$

with $p + \frac{1}{2}q > 1$. We let the diameter $\|\Pi\| \rightarrow 0^+$, let the partition be finer. Use the fact that $(dW(u))^2 = du$ and $(dt)^2 = 0$, $dt dW(t) = 0$, we obtain that the above sum will tend to

$$\int_0^T f_t(u, W(u))du + \int_0^T f_x(u, W(u))dW(u) + \frac{1}{2} \int_0^T f_{xx}(u, W(u))du.$$

And we obtain the Itô formula for Brownian motion

$$f(T, W(T)) = f(0, W(0)) + \int_0^T f_t(u, W(u))du + \int_0^T f_x(u, W(u))dW(u) + \frac{1}{2} \int_0^T f_{xx}(u, W(u))du, \quad (4)$$

or in other words,

$$df(u, W(u)) = \left\{ f_t(u, W(u)) + \frac{1}{2}f_{xx}(u, W(u)) \right\} dt + f_x(u, W(u))dW(u). \quad (5)$$

¹To control the increments of Brownian motion in each interval $[t_{i-1}, t_i]$, one can let the partition Π to be random, i.e., let the partition points t_i to be stopping times.

2 Formula for Itô Process

An Itô process is defined with a stochastic integral with respect to Brownian motion. In other words, an Itô process $\{X(t)\}_{t \geq 0}$ has the following form:

$$X(t) = X(0) + \int_0^t \Delta(u) dW(u) + \int_0^t \Theta(u) du, \quad (6)$$

where $X(0)$ is deterministic initial value, $\{\Delta(u)\}$ and $\{\Theta(u)\}$ are stochastic processes adapted to the filtration of the Brownian motion $\{W(u)\}$.

The differential form of (5) is simpler:

$$dX(t) = \Delta(t) dW(t) + \Theta(t) dt, \quad (7)$$

from which we obtain that the infinitesimal increment of the quadratic variation of X is

$$(dX(t))^2 = \Delta^2(t) dt. \quad (8)$$

Because of (6), we can legally define stochastic integrals with respect to the Itô process: for any adapted process $\{\Gamma(t)\}$, we can define

$$\int_0^t \Gamma(u) dX(u) := \int_0^t \Gamma(u) \Delta(u) dW(u) + \int_0^t \Gamma(u) \Theta(u) du. \quad (9)$$

Now we can mimic what we have done for Brownian motion: given a good function $f(t, x)$, we have

$$\begin{aligned} & df(u, X(u)) \\ &= f_t(u, X(u)) du + f_x(u, X(u)) dX(u) + \frac{1}{2} f_{xx}(u, X(u)) (dX(u))^2 \\ &= \left\{ f_t(u, X(u)) + \Theta(u) f_x(u, X(u)) + \frac{1}{2} \Delta^2(u) f_{xx}(u, X(u)) \right\} dt + \Delta(u) f_x(u, X(u)) dW(u). \end{aligned} \quad (10)$$

This is Itô formula for general Itô processes.

3 Examples

3.1 Generalized geometric Brownian motion

Let $\{W(t)\}_{t \geq 0}$ be a standard Brownian motion starting at zero. Let $\{\alpha(t)\}_{t \geq 0}$ and $\{\sigma(t)\}_{t \geq 0}$ be adapted processes. Define the Itô process

$$X(t) = \int_0^t \sigma(u) dW(u) + \int_0^t \left(\alpha(u) - \frac{1}{2} \sigma^2(u) \right) du. \quad (11)$$

Then we have

$$dX(u) = \sigma(u) dW(u) + \left(\alpha(u) - \frac{1}{2} \sigma^2(u) \right) du. \quad (12)$$

The generalized geometric Brownian motion is given by

$$S(t) = S(0) e^{X(t)} = S(0) \exp \left\{ \int_0^t \sigma(u) dW(u) + \int_0^t \left(\alpha(u) - \frac{1}{2} \sigma^2(u) \right) du \right\}, \quad (13)$$

for a positive constant $S(0)$. Process (13) is usually used to model asset price. Let us investigate its dynamics more closely. Let $f(x) = S(0) e^x$, then we have $S(t) = f(X(t))$. Applying Itô formula (10) we have that

$$\begin{aligned} dS(t) &= df(X(t)) = f'(X(t)) dX(t) + \frac{1}{2} f''(X(t)) (dX(t))^2 \\ &= S(t) dX(t) + \frac{1}{2} S(t) (dX(t))^2 = \alpha(t) S(t) dt + \sigma(t) S(t) dW(t). \end{aligned} \quad (14)$$

We call $\alpha(t)$ and $\sigma(t)$ the instantaneous mean rate of return and volatility. If $\alpha(t) = 0$ for all $t > 0$, then $\{S(t)\}$ is a martingale, since $S(t)$ is a stochastic integral w.r.t Brownian motion:

$$S(t) = S(0) + \int_0^t \sigma(u) S(u) dW(u). \quad (15)$$

Similarly, if $\alpha(t) = r$ for $t > 0$, where r is the risk-free interest rate, then the discounted asset price process $\{e^{-rt} S(t)\}$ is a martingale.

3.2 Itô integral of a deterministic integrand

Let $\{\sigma(t)\}_{t \geq 0}$ be a deterministic process, and $\{W(t)\}_{t \geq 0}$ be a standard Brownian motion starting at zero. Define

$$I(t) = \int_0^t \sigma(u) dW(u). \quad (16)$$

In one homework problem we have shown that, if $\{\sigma(t)\}$ is a simple integrand, then (16) is normally distributed with zero mean and variance $\int_0^t \sigma^2(u) du$. We prove this is true for general $\{\sigma(t)\}$.

Following the discussion in last part, for any $\lambda \in \mathbb{R}$, we have a martingale

$$S(t) = \exp \left\{ \lambda \int_0^t \sigma(u) dW(u) - \frac{1}{2} \lambda^2 \int_0^t \sigma^2(u) du \right\}, \quad t \geq 0. \quad (17)$$

Therefore the expectation does not change with time:

$$1 = S(0) = e^{-\frac{1}{2} \lambda^2 \int_0^t \sigma^2(u) du} E \exp \left\{ \lambda \int_0^t \sigma(u) dW(u) \right\}, \quad (18)$$

from which we obtain the moment-generating function of random variable $I(t)$:

$$E e^{\lambda I(t)} = e^{\frac{1}{2} \lambda^2 \int_0^t \sigma^2(u) du}. \quad (19)$$

Recall that a normal random variable with mean μ and variance σ^2 has a moment-generating function of $e^{\mu t + \frac{1}{2} \sigma^2 t^2}$, we thus conclude that $I(t)$ is $N(0, \int_0^t \sigma^2(u) du)$.

3.3 Vasicek interest rate model

Let $\{W(t)\}_{t \geq 0}$ be a standard Brownian motion starting at zero. For positive parameters θ, κ, σ , the Vasicek interest rate process (aka the Ornstein-Uhlenbeck process) is defined as

$$dR(t) = \kappa(\theta - R(t))dt + \sigma dW(t), \quad R(0) > 0. \quad (20)$$

The solution to equation (20) admits a closed form solution:

$$R(t) = e^{-\kappa t} R(0) + \theta(1 - e^{-\kappa t}) + \sigma e^{-\kappa t} \int_0^t e^{\kappa u} dW(u). \quad (21)$$

To verify (21) is the solution to (20), consider function

$$f(t, x) = e^{-\kappa t} R(0) + \theta(1 - e^{-\kappa t}) + \sigma e^{-\kappa t} x, \quad (22)$$

and $X(t) = \int_0^t e^{\kappa u} dW(u)$. Applying Itô formula (10) to $f(t, X(t))$ we have that

$$\begin{aligned} df(u, X(u)) &= f_t(u, X(u))du + f_x(u, X(u))dX(u) + \frac{1}{2} f_{xx}(u, X(u))(dX(u))^2 \\ &= \kappa(\theta - f(u, X(u)))du + \sigma dW(u). \end{aligned} \quad (23)$$

Moreover, $f(0, X(0)) = R(0)$, so $R(t) = f(t, X(t))$ is the solution to (20).

From last part, the representation (21) implies that $R(t)$ is normally distributed with mean $e^{-\kappa t} R(0) + \theta(1 - e^{-\kappa t})$, and variance

$$\sigma^2 e^{-2\kappa t} \int_0^t e^{2\kappa u} du = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa t}). \quad (24)$$

Vasicek model has the desirable property that the interest rate is mean-reverting. In the model, θ is the long-term mean. A drawback of Vasicek model is that the rate can go negative.

3.4 Cox-Ingersoll-Ross (CIR) interest rate model

Let $\{W(t)\}_{t \geq 0}$ be a standard Brownian motion starting at zero. For positive parameters θ, κ, σ , the CIR interest rate process (aka, the square-root process²) is defined as

$$dR(t) = \kappa(\theta - R(t))dt + \sigma\sqrt{R(t)}dW(t), \quad R(0) > 0. \quad (25)$$

Similar as Vasicek model, the CIR interest rate process is mean-reverting. However, there is no closed-form solution to (25), and the process $R(t)$ is always non-negative³.

We can use Itô formula to obtain the mean and the variance of $R(t)$. To do so, let $f(t, x) = e^{\kappa t}x$, then

$$\begin{aligned} d(e^{\kappa u}R(u)) &= f(u, R(u)) = f_t(u, R(u))du + f_x(u, R(u))dR(u) + \frac{1}{2}f_{xx}(u, R(u))(dR(u))^2 \\ &= \kappa e^{\kappa u}R(u)du + \kappa e^{\kappa u}(\theta - R(u))du + e^{\kappa u}\sigma\sqrt{R(u)}dW(u) \\ &= \kappa\theta e^{\kappa u}du + e^{\kappa u}\sigma\sqrt{R(u)}dW(u), \end{aligned} \quad (26)$$

from which we obtain that

$$\begin{aligned} e^{\kappa t}R(t) &= R(0) + \theta \int_0^t \kappa e^{\kappa u}du + \sigma \int_0^t e^{\kappa u}\sqrt{R(u)}dW(u) \\ &= R(0) + \theta(e^{\kappa t} - 1) + \sigma \int_0^t e^{\kappa u}\sqrt{R(u)}dW(u). \end{aligned} \quad (27)$$

So we have

$$e^{\kappa t}E(R(t)) = R(0) + \theta(e^{\kappa t} - 1), \quad (28)$$

or, equivalently,

$$ER(t) = e^{-\kappa t}R(0) + \theta(1 - e^{-\kappa t}). \quad (29)$$

To compute the variance of $R(t)$, we need $ER^2(t)$. To obtain it, set $X(t) = e^{\kappa t}R(t)$, then (26) means

$$dX(u) = \theta\kappa e^{\kappa u}du + \sigma e^{\frac{\kappa u}{2}}\sqrt{X(u)}dW(u). \quad (30)$$

Using Itô formula and (30) we obtain that

$$\begin{aligned} d(X^2(u)) &= 2X(u)dX(u) + (dX(u))^2 \\ &= 2\theta\kappa e^{\kappa u}X(u)du + 2\sigma e^{\frac{\kappa u}{2}}X^{\frac{3}{2}}(u)dW(u) + \sigma^2 e^{\kappa u}X(u)du. \end{aligned} \quad (31)$$

In other words,

$$X^2(t) = X^2(0) + (2\theta\kappa + \sigma^2) \int_0^t e^{\kappa u}X(u)du + 2\sigma \int_0^t e^{\frac{\kappa u}{2}}X^{\frac{3}{2}}(u)dW(u). \quad (32)$$

Taking expectation and interchanging the order of expectation and integration we obtain that

$$\begin{aligned} EX^2(t) &= X^2(0) + (2\theta\kappa + \sigma^2) \int_0^t e^{\kappa u}EX(u)du \\ &= R^2(0) + (2\theta\kappa + \sigma^2) \int_0^t e^{\kappa u}(R(0) + \theta(e^{\kappa u} - 1))du \\ &= R^2(0) + \left(2\theta + \frac{\sigma^2}{\kappa}\right)(R(0) - \theta)(e^{\kappa t} - 1) + \left(\theta + \frac{\sigma^2}{2\kappa}\right)\theta(e^{2\kappa t} - 1). \end{aligned} \quad (33)$$

Thus,

$$ER^2(t) = e^{-2\kappa t}R^2(0) + \left(2\theta + \frac{\sigma^2}{\kappa}\right)(R(0) - \theta)(e^{-\kappa t} - e^{-2\kappa t}) + \left(\theta + \frac{\sigma^2}{2\kappa}\right)\theta(1 - e^{-2\kappa t}). \quad (34)$$

²This process is also used in Heston's stochastic volatility model.

³CIR process always stays positive if $2\kappa\theta > \sigma^2$.

In conclusion, we have

$$\begin{aligned}\mathrm{Var}(R(t)) &= ER^2(t) - (ER(t))^2 \\ &= \frac{\sigma^2}{\kappa}R(0)(e^{-\kappa t} - e^{-2\kappa t}) + \frac{\theta\sigma^2}{2\kappa}(1 - e^{-\kappa t})^2.\end{aligned}\tag{35}$$

In particular,

$$\lim_{t \rightarrow \infty} \mathrm{Var}(R(t)) = \frac{\theta\sigma^2}{2\kappa}.\tag{36}$$